- If a bond's price is known, the bond's implied return can be computed using the bond's price and its promised future cash flows.
- A stock's required return can be estimated given the stock's current price and assumptions about its expected future dividends and growth rates.
- A stock's implied dividend growth rate can be estimated given the stock's current price and assumptions about its expected future dividends and required return.
- If valuing two (or more) cash flow streams, the cash flow additivity principle allows for the cash flow streams to be compared (as long as the cash flows occur at the same point in time).
- Application of cash flow additivity allows for confirmation that asset prices are the same for economically equivalent assets (even if the assets have differing cash flow streams).
- Several real-world applications of cash flow additivity are used to illustrate no-arbitrage pricing.

2 TIME VALUE OF MONEY IN FIXED INCOME AND EQUITY

☐ calculate and interpret the present value (PV) of fixed-income and equity instruments based on expected future cash flows

The timing of cash flows associated with financial instruments affects their value, with cash inflows valued more highly the sooner they are received. The time value of money represents the trade-off between cash flows received today versus those received on a future date, allowing the comparison of the current or present value of one or more cash flows to those received at different times in the future. This difference is based upon an appropriate discount rate r as shown in the prior learning module, which varies based upon the type of instrument and the timing and riskiness of expected cash flows.

In general, the relationship between a current or present value (PV) and future value (FV) of a cash flow, where r is the stated discount rate per period and t is the number of compounding periods, is as follows:

$$\mathrm{FV}_t = \mathrm{PV}(1 + r)^t. \tag{1}$$

If the number of compounding periods t is very large, that is, $t \to \infty$, we compound the initial cash flow on a continuous basis as follows:

$$\mathrm{FV}_t = \mathrm{PV}e^{r\,t}. \tag{2}$$

Conversely, present values can be expressed in future value terms, which requires recasting Equation 1 as follows:

$$\begin{aligned} \mathrm{FV}_t &= \mathrm{PV}(1 + r)^t \\ \mathrm{PV} &= \mathrm{FV}_t\left[\frac{1}{(1 + r)^t}\right]. \\ \mathrm{PV} &= \mathrm{FV}_t(1 + r)^{-t} \end{aligned} \tag{3}$$

The continuous time equivalent expression of Equation 3 is as follows: